DBMS (CSD 202): Quiz 1 **15-02-2019**

Time: 45 minutes Marks: 21

User

Id	Name	Age	Gender	OccupationId	CityId
1	John	25	Male	1	3
2	Sara	20	Female	3	4
3	Victor	31	Male	2	5
4	Jane	27	Female	1	3

Occupation

OccupationId	OccupationName
1	Software Engineer
2	Accountant
3	Pharmacist
4	Library Assistant

City

CityId	CityName
1	Halifax
2	Calgary
3	Boston
4	New York
5	Toronto

1. Solve the following relational expressions for above relations.

a. $P_{Name}(R_{Age>25}(User))$

b. $R_{Id>2 \vee Age!=31}(User)$

c. $R_{User.OccupationId=Occupation.OccupationId}(User \times Occupation)$

d. $User \bowtie Occupation \bowtie City$

e. $P_{Name,Gender}(R_{CityName="Boston"}(User \bowtie City))$

f. $P_{Name,\ CityName}\ (User \bowtie_{\sqsubset} City)$

g. User * City

Where R and P represents select and project operations respectively.

Write the relational algebra expression for following queries:

h. For each city, list the city name and number of users live in that city.

i. Find the names of all the female Software Engineers living in 'Boston'.

(14)

DBMS (CSD 202): Quiz 1 **15-02-2019**

Time: 45 minutes Marks: 21

Answers

a. $P_{Name}(R_{Age>25}(User))$

Name
Victor
Jane

```
SELECT Name
FROM User
WHERE Age > 25;
```

b. $R_{Id>2 \vee Age!=31}(User)$

Id	Name	Age	Gender	OccupationId	CityId
1	John	25	Male	1	3
2	Sara	20	Female	3	4
3	Victor	31	Male	2	5
4	Jane	27	Female	1	3

```
SELECT *
FROM User
WHERE id>2 OR Age != 31;
```

01intro [Protected View] - PowerPoi

c. $R_{User.OccupationId=Occupation.OccupationId}(User \times Occupation)$

Id	Name	Age	Gender	OccupationId	CityId	OccupationId	OccupationName
1	John	25	Male	1	3	1	Software Engineer
2	Sara	20	Female	3	4	3	Pharmacist
3	Victor	31	Male	2	5	2	Accountant
4	Jane	27	Female	1	3	1	Software Engineer

```
SELECT *
FROM User u, Occupation o
WHERE u.OccupationId = o.OccupationId;
```

d. User ⋈ Occupation ⋈ City

CityId	OccupationId	Id	Name	Age	Gender	OccupationName	CityName
3	1	1	John	25	Male	Software Engineer	Boston
4	3	2	Sara	20	Female	Pharmacist	New York
5	2	3	Victor	31	Male	Accountant	Toronto
3	1	4	Jane	27	Female	Software Engineer	Boston

```
SELECT *
FROM User NATURAL JOIN Occupation NATURAL JOIN City;
```

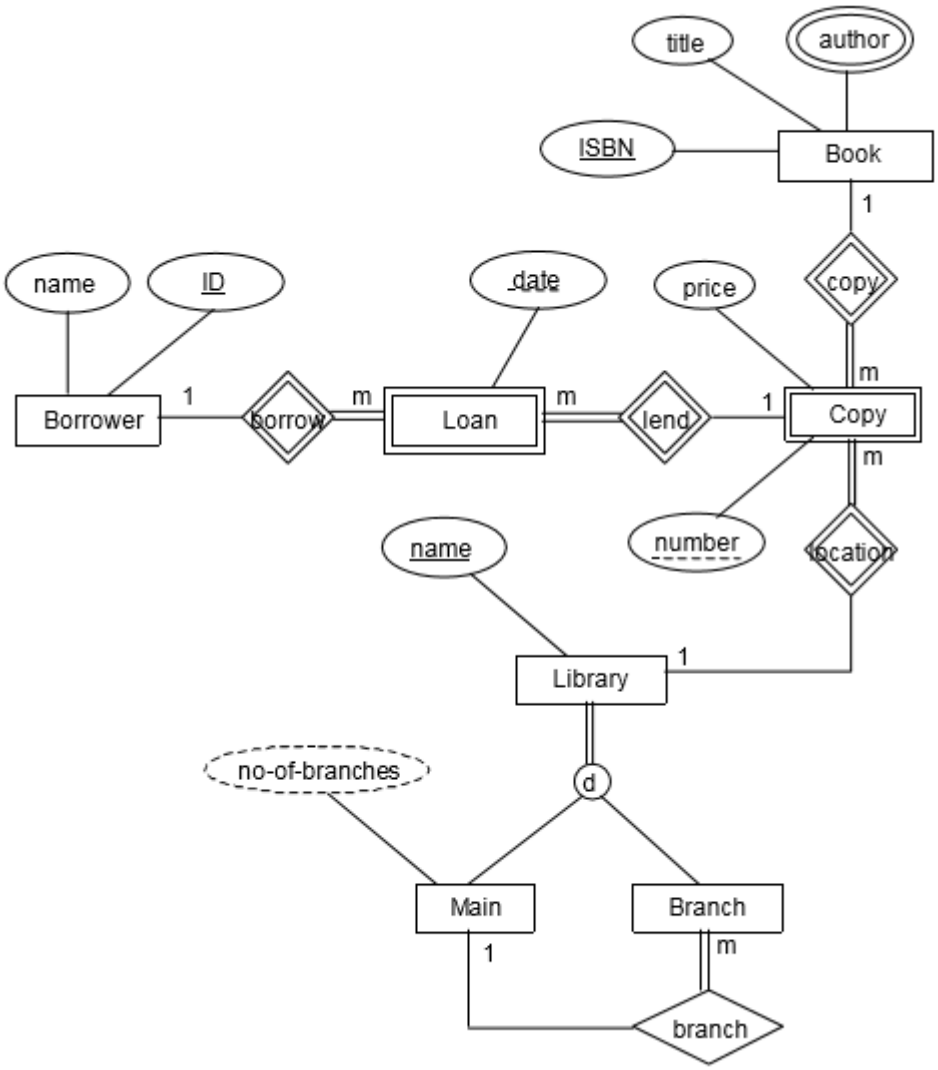
e. $P_{Name,Gender}(R_{CityName="Boston"}(User \bowtie City))$

Name	Gender
John	Male
Jane	Female

```
SELECT Name, Gender
FROM User NATURAL JOIN City
WHERE CityName = ”Boston”;
```

2. A library service wants to create a database to store details of its libraries, books and borrowers. Details include the following:

A book has a unique ISBN number, a title and one or more authors. The library service may own several copies of a given book, each of which is located in one of the service’s libraries. A given library contains many books, and in order to distinguish different copies of the same book a library assigns a different copy-number to each of its copies of a given book; the price that was paid for each copy is also recorded. Every library has a unique name and is either a main library or a branch library. A main library may have zero or more branch libraries and every branch library is a branch of exactly one main library. A borrower has a name and a unique ID code. A borrower can have many books on loan, but each copy of a book can only be on loan to one borrower. A borrower could borrow

the same book on several occasions, but it is assumed that each such loan will take place on a different date.

Draw the EER diagram for the library service. **(7)**